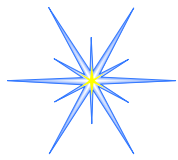


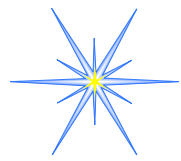
Embedded, Real-Time and Wireless Systems

Professor Jack Stankovic
Department of Computer Science
University of Virginia
June 2, 2005

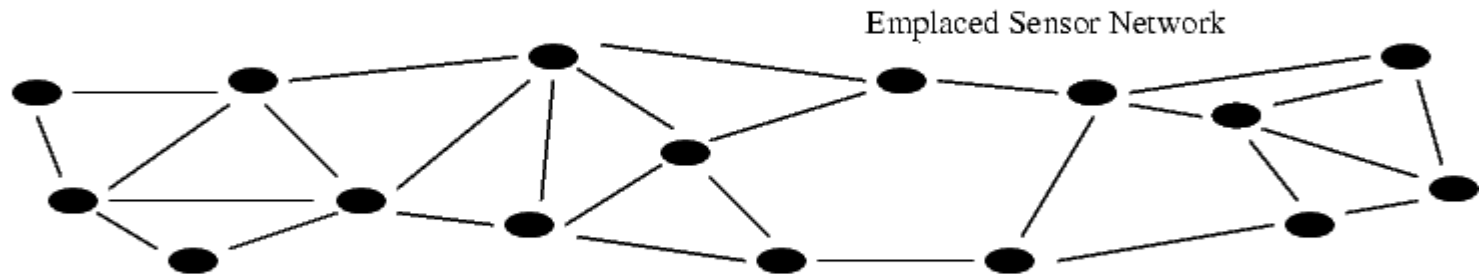


Outline

- Embedded, Real-Time Wireless Infrastructure
- Map to Smart Living Spaces, e.g., Assisted Living/Nursing Homes
 - Key Issue: Many Stakeholders
- Advantages
- Research Questions



Medical System Architecture



Wireless

Berkeley Motes

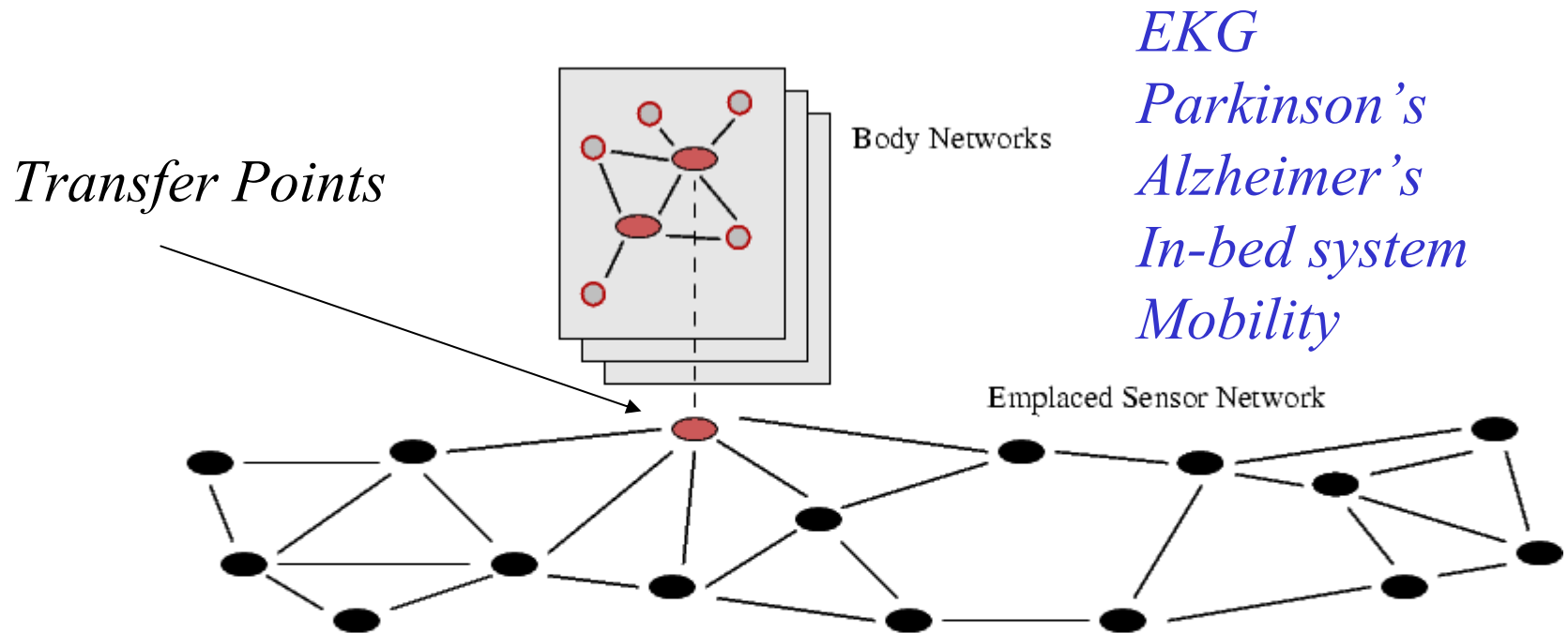
Environmental Sensors

Location

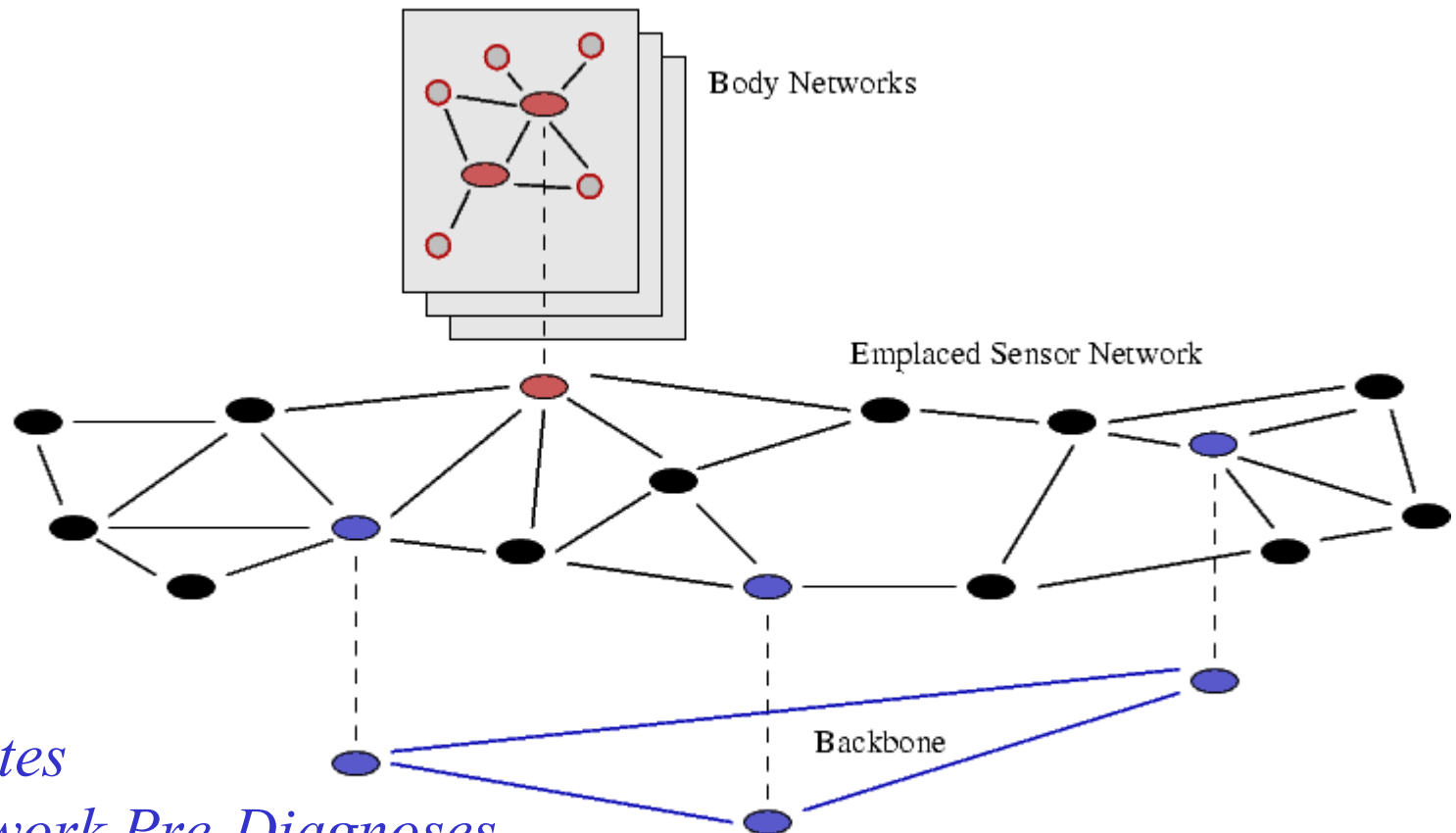
Identify Individuals

Ad hoc, flexible, easy to rearrange, etc.

Medical System Architecture



Medical System Architecture

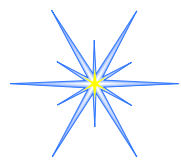


Stargates

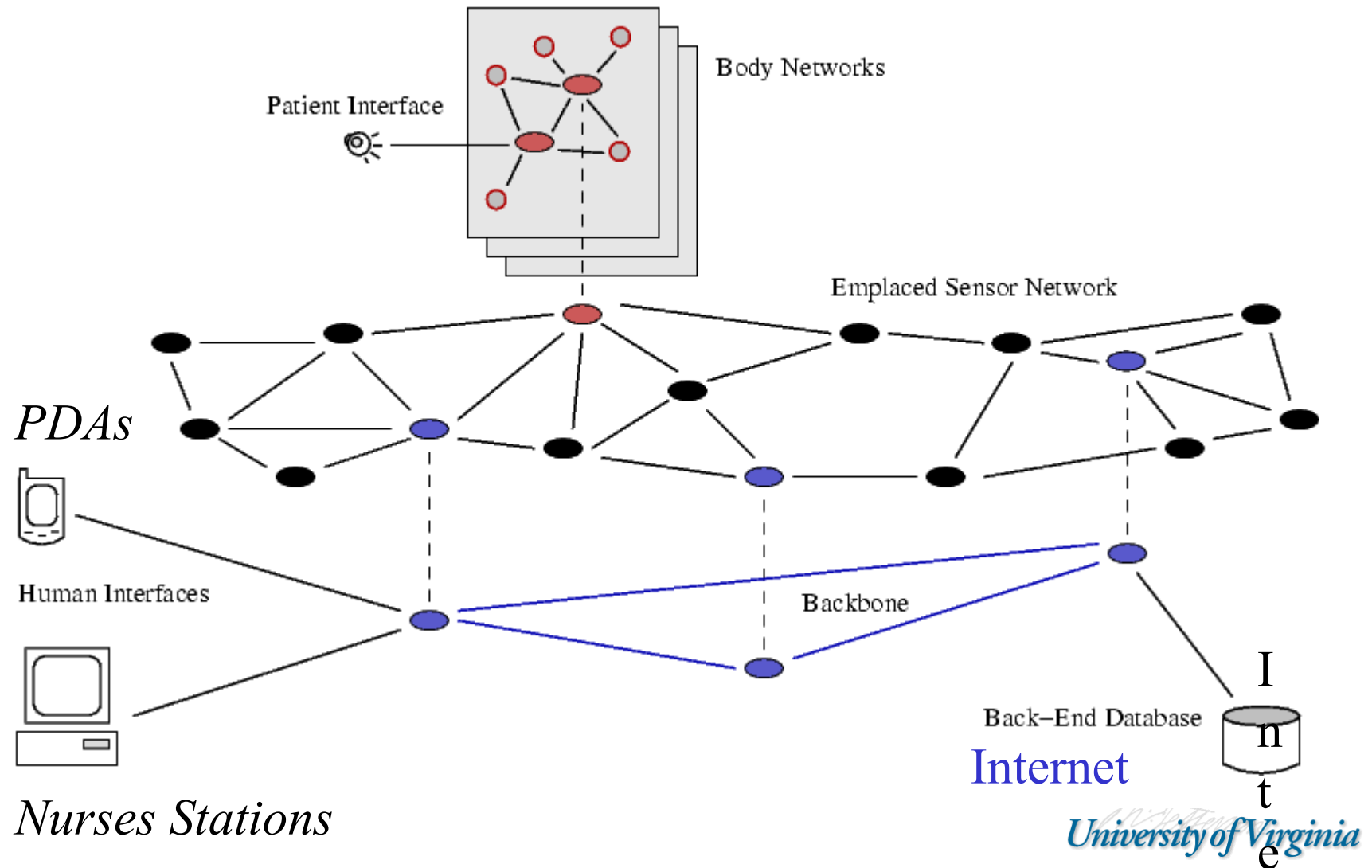
In-network Pre-Diagnoses

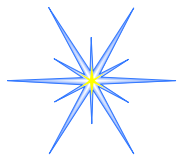
Storage

Video/Acoustic Streaming

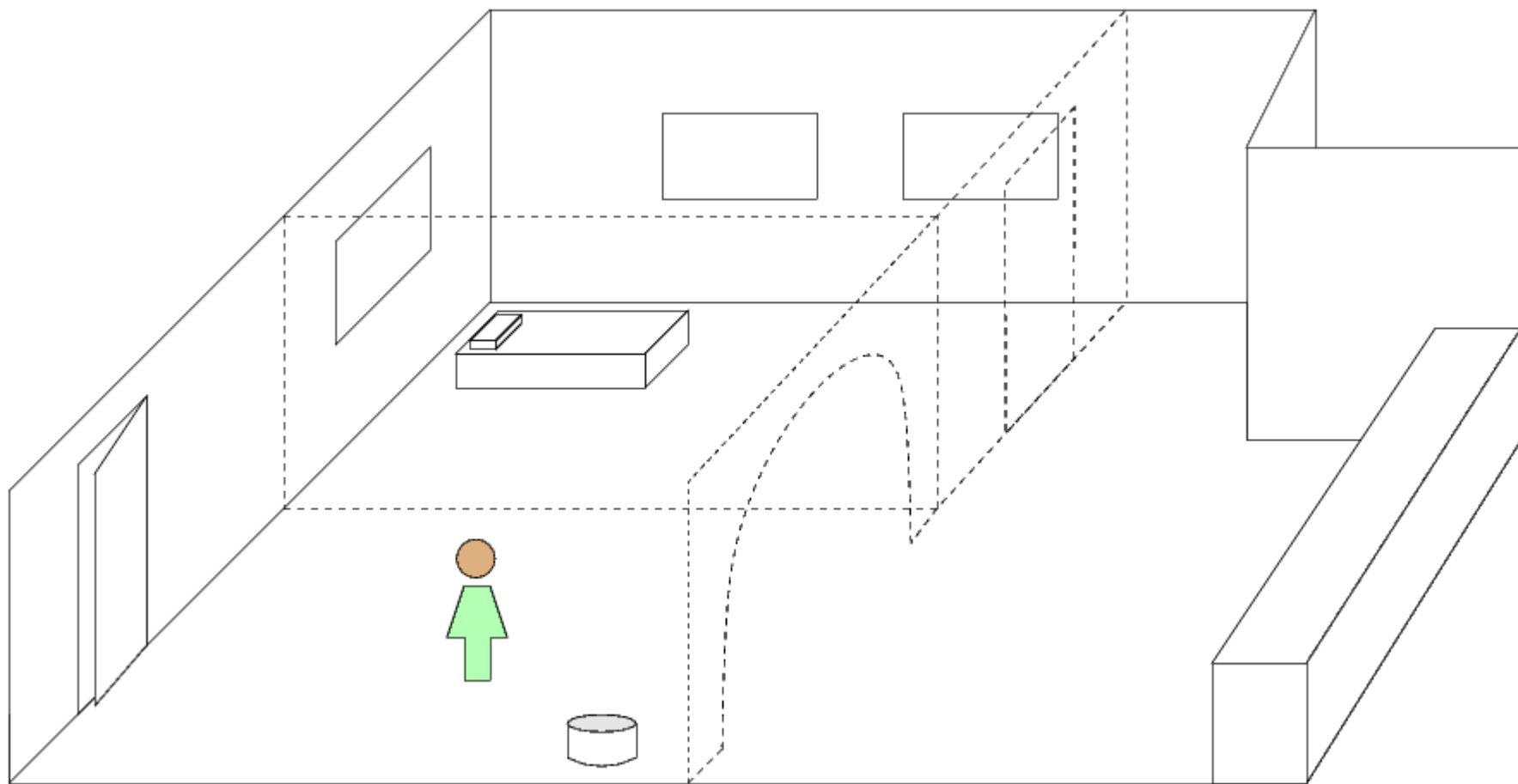


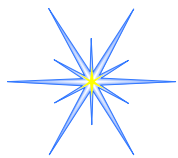
Medical System Architecture



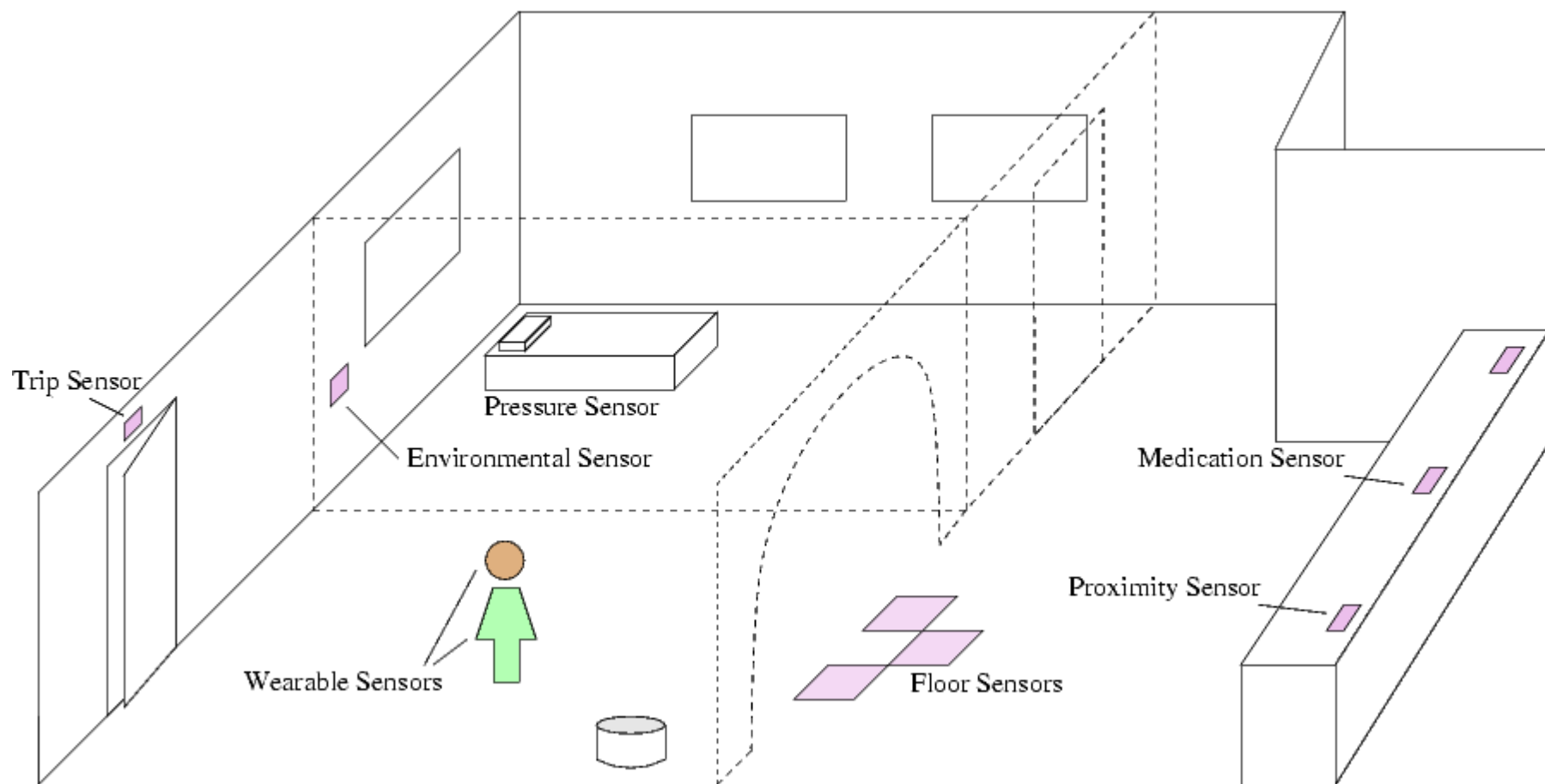


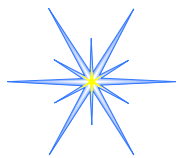
Smart Living Health Spaces



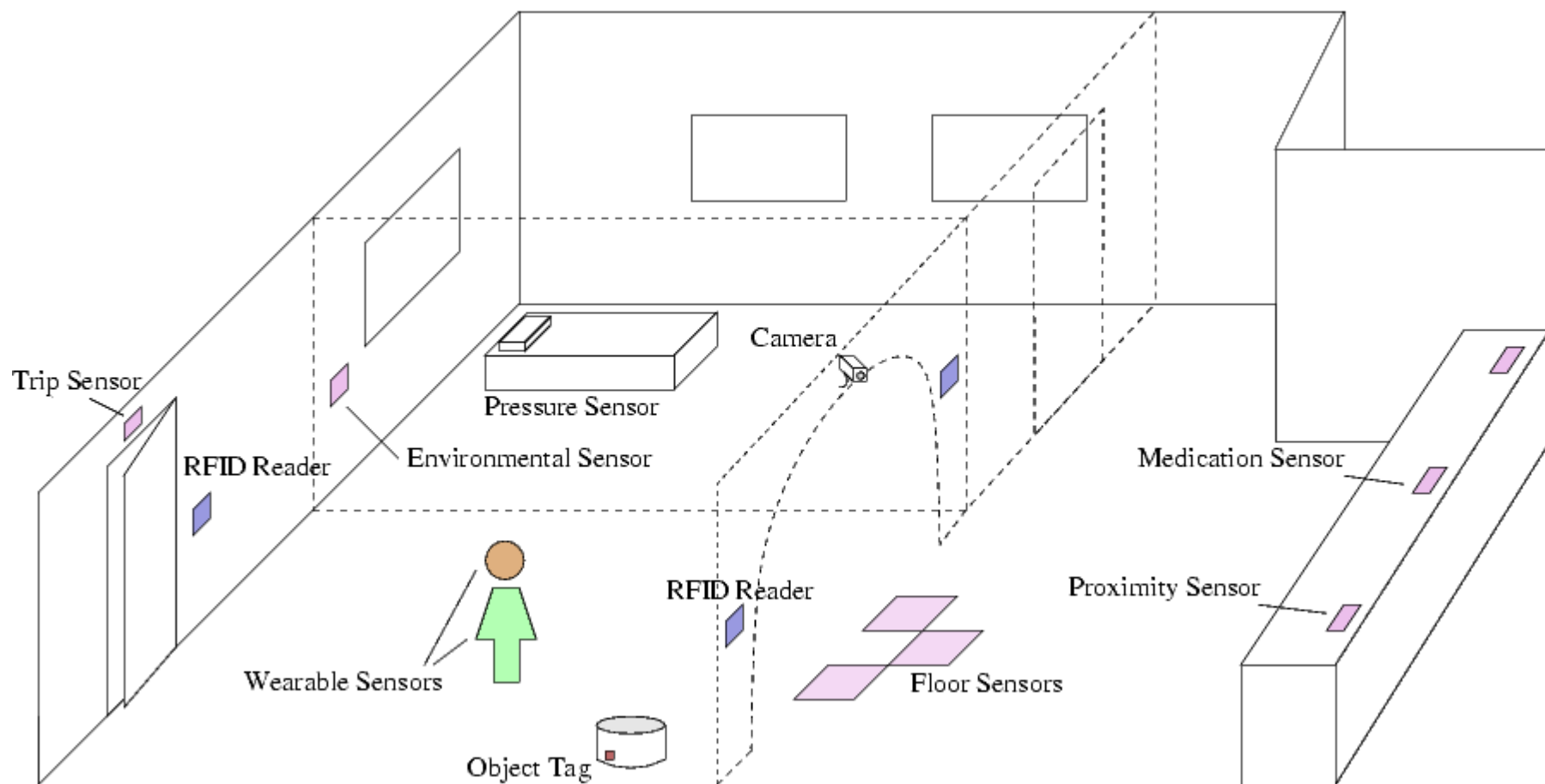


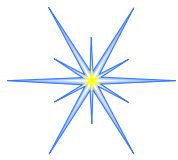
Smart Living Health Spaces



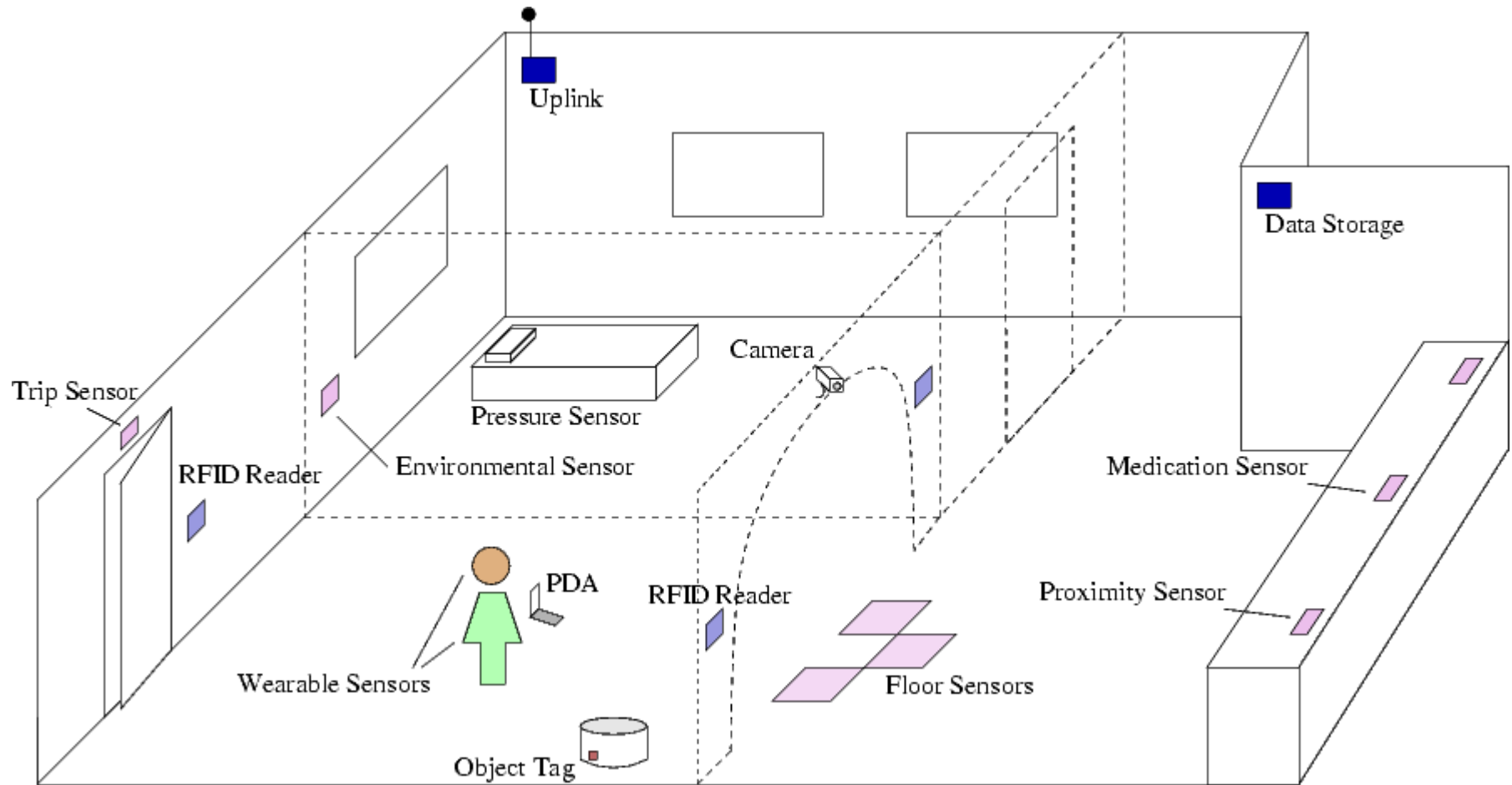


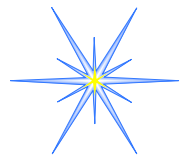
Smart Living Health Spaces





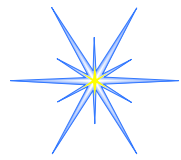
Smart Living Health Spaces





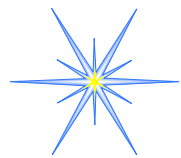
Many Stakeholders

- Patients
- Patients family and friends
- Doctor - what advantages for them in treating patients
- Nurse
- Technician
- Orderly
- Admin
- Social Worker



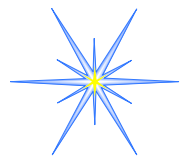
Advantages

- Tailored to Patient Health
 - Improved Health Care
- Continuous Monitoring
 - More natural settings
 - More complete
- Improve Quality of Life
 - Preserve Dignity
 - More freedom of movement
- Collect Data for Longitudinal Studies
- Rapid and Flexible Deployment
- Easy to Modify over Time
- Protects Privacy



Research Questions

- Flexible and Dynamic Privacy
- Security
- Form factors for sensor nodes
 - Unobtrusiveness
- Mobility
 - Routing for 2 mobile end points
- Localization
- In-network preliminary diagnoses
- Define and meet real-time requirements including alarms



Research Questions

- Body network operating system
- Support for in-network storage and queries
- Robustness/Reliability
 - Wireless networking
- Power Management
- HCI
- Satisfying "n" stakeholders
- N-people (monitoring, tracking, distinguishing, etc.)